

PROJECT TECHNICAL REPORT

AI-Powered Steel Defect Inspection and Root Cause Analysis System

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GitHub: <https://github.com/baheti777/ai-steel-defect-inspection-system>

Live Demo: <https://huggingface.co/spaces/bhuviiiiiii/steel-defect-inspector>

01. *Executive Summary*

This report documents an AI-powered steel surface defect inspection and root cause analysis system developed by me, Bhuvi Baheti. The project was inspired by recent research on explainable AI for steel defect inspection, particularly the work of Madhavan et al. (2024). Building upon these ideas, the present work extends defect classification through GradCAM-based explainability, defect severity assessment, metallurgical root-cause analysis, RPN-based risk scoring, and corrective action recommendations.

The system combines deep learning, explainable AI, and metallurgical domain knowledge to deliver an end-to-end quality inspection pipeline. Using an EfficientNet-B3 model trained on the NEU Steel Surface Defect Dataset, the system classifies six common steel surface defects and provides interpretable insights to support quality engineers. The complete solution is deployed as an interactive web application using Gradio and Hugging Face Spaces.

Experimental evaluation achieved 100% validation accuracy on the NEU-DET benchmark dataset, demonstrating the potential of AI-driven inspection systems for intelligent quality control and Industry 4.0 manufacturing applications.

02. *Problem Statement & Industrial Context*

Steel surface defects are a critical quality concern in modern manufacturing. They lead to product rejection, increased production costs, customer dissatisfaction, and downstream structural failures. In rolling mills, automotive sheet production, and structural steel plants, surface quality directly determines product viability.

Limitations of Conventional Approaches

Manual visual inspection is slow, subjective, and fatigue-prone — unsuitable for high-throughput production lines. Existing automated AI systems typically address only defect classification, leaving quality engineers without the causal insights needed for process correction. This creates a critical gap between defect detection and actionable remediation.

The Industry 4.0 Opportunity

With the rise of smart manufacturing, there is a growing demand for intelligent inspection systems that not only detect defects, but explain their origin, quantify their severity, and prescribe corrective actions. This project directly addresses that unmet need.

03. *Project Objectives*

- Develop a high-accuracy deep learning model for steel surface defect classification across 6 categories.
- Implement GradCAM-based explainability to visualize which surface regions drive model predictions.
- Estimate defect severity using activated defect area from GradCAM heatmaps.
- Build a metallurgical knowledge base linking each defect class to probable process-level root causes.
- Implement RPN (Risk Priority Number)-based industrial risk scoring inspired by FMEA methodology.
- Provide structured corrective action recommendations for each defect type.
- Deploy the complete system as an interactive web application accessible via Hugging Face Spaces.

04. Dataset

The project uses the NEU Steel Surface Defect Dataset (NEU-DET) — a widely cited benchmark in industrial defect detection research — sourced from Kaggle. This dataset provides a realistic and challenging benchmark for steel surface inspection.

Property	Detail	Notes
Total Images	1,800	Balanced across 6 classes
Images per Class	300	Equal class distribution
Image Format	Grayscale JPEG	200 × 200 pixels
Defect Classes	6	Industry-representative
Benchmark Status	Widely Used	Published research standard

05. Steel Defect Classes

The system classifies steel surface images into the following six defect categories, each linked to distinct manufacturing process anomalies:

Defect Class	Description	RPN Score
Crazing	Fine crack network from thermal stresses	8/10
Inclusion	Non-metallic particles embedded in steel matrix	10/10
Patches	Irregular scale or discoloration defects	5/10

Defect Class	Description	RPN Score
Pitted Surface	Surface craters from corrosion or casting porosity	7/10
Rolled-in Scale	Oxide scale folded into steel during hot rolling	9/10
Scratches	Linear marks from mechanical contact or abrasion	4/10

06. *System Architecture & Workflow*

End-to-End Pipeline

The system processes steel surface images through a sequential analytical pipeline:

1. Input Image Acquisition — Steel surface image uploaded via Gradio interface.
2. Preprocessing — Resize to 224×224 px, normalize using ImageNet statistics.
3. EfficientNet-B3 Inference — Deep learning model produces class probabilities.
4. GradCAM Visualization — Generates spatial activation heatmap overlaid on original image.
5. Defect Area Estimation — Activations above 0.5 threshold quantify defect extent.
6. Severity Classification — Critical (>30%), Moderate (>15%), or Minor (<15%).
7. Root Cause Analysis — Knowledge base maps defect to probable process causes.
8. RPN Risk Scoring — Industrial risk prioritization (1–10 scale).
9. Corrective Action Generation — Process-specific recommendations provided.
10. Report Output — Comprehensive metallurgical analysis report generated.

07. *Deep Learning Model*

Architecture: EfficientNet-B3

EfficientNet-B3 was selected as the backbone architecture based on its superior performance-to-parameter ratio. The model achieves state-of-the-art accuracy on image classification benchmarks while remaining computationally efficient — critical for deployment in industrial edge environments.

Model Parameter	Specification
Base Architecture	EfficientNet-B3 (torchvision)
Pre-training	ImageNet1K (transfer learning)
Input Resolution	224 × 224 pixels

Model Parameter	Specification
Custom Classifier Head	Dropout(0.4) → Linear(1280→512) → ReLU → Dropout(0.3) → Linear(512→6)
Output Classes	6 defect categories
Framework	PyTorch
Deployment	Gradio + Hugging Face Spaces

Why EfficientNet-B3?

- Compound scaling of depth, width, and resolution delivers superior accuracy over ResNet and VGG at lower parameter counts.
- Transfer learning from ImageNet provides robust low-level feature extraction, particularly valuable with limited domain-specific training data.
- Efficient enough for CPU-based inference, enabling deployment without dedicated GPU infrastructure.
- Well-suited for Hugging Face Spaces free-tier hosting constraints.

08. *Explainability: GradCAM*

A fundamental limitation of neural networks in industrial applications is their "black box" nature — providing predictions without justification. GradCAM (Gradient-weighted Class Activation Mapping) solves this by generating spatial heatmaps that highlight the image regions most influential in the model's classification decision.

Technical Implementation

- Forward hooks capture feature map activations from the final convolutional layer (`model.features[-1]`).
- Backward hooks capture gradient signals flowing back from the predicted class logit.
- Activations are weighted by spatially-averaged gradients, then ReLU-activated to retain positive influences.
- The resulting heatmap is resized to input resolution and overlaid as a jet colormap on the original image.

Industrial Value

GradCAM overlays allow quality engineers to visually verify that the model is focusing on actual defect regions — not background artifacts. This transparency is essential for regulatory compliance and operator trust in AI-assisted inspection systems.

09. *Defect Severity Assessment*

Unlike conventional systems that use prediction confidence as a proxy for severity, this system employs a physically meaningful severity metric derived from the GradCAM activation map.

Methodology

The fraction of image pixels with GradCAM activation above 0.5 threshold is computed as the defect area percentage. This approximates the spatial extent of the defect on the steel surface.

Severity Level	Defect Area Threshold	Indicator
Critical	> 30% of surface	Red — Immediate action required
Moderate	15% – 30% of surface	Orange — Inspection recommended
Minor	< 15% of surface	Green — Monitor and log

10. Metallurgical Root Cause Analysis

A structured metallurgical knowledge base maps each defect class to probable manufacturing process failures. This bridges the gap between defect detection and corrective action — the critical missing link in conventional inspection systems.

Knowledge Base Structure

For each of the 6 defect classes, the knowledge base encodes:

- Defect Description — Physical characteristics and visual appearance.
- Probable Root Causes — 3 process-level causes based on metallurgical engineering principles.
- Process Parameters to Check — Specific inspection points for quality engineers.
- Corrective Action Recommendations — Actionable process adjustments to prevent recurrence.

Example: Rolled-In Scale

Root Causes	Inadequate descaling before hot rolling; excessive furnace scale formation; blocked descaler nozzles.
Process Checks	Descaler nozzle condition and pressure logs; reheating furnace temperature and soaking time; inter-pass descaling between rolling stands.
Corrective Actions	Increase descaling pressure to ≥180 bar; reduce furnace soaking time at peak temperature; schedule regular nozzle maintenance.

11. *RPN-Based Industrial Risk Assessment*

The system incorporates Risk Priority Numbers (RPN) inspired by industry-standard FMEA (Failure Mode and Effects Analysis) methodology. RPN scores enable quality engineers to prioritize defect types by their potential impact on product integrity and production economics.

Defect Type	RPN Score (1-10)	Risk Level
Inclusion	10 / 10	Critical — Structural integrity risk
Rolled-in Scale	9 / 10	Very High — Surface quality failure
Crazing	8 / 10	High — Fatigue crack initiation
Pitted Surface	7 / 10	Significant — Corrosion acceleration
Patches	5 / 10	Moderate — Aesthetic and coating issues
Scratches	4 / 10	Low-Moderate — Handling artifact

12. *Web Application & Deployment*

The complete inspection pipeline is deployed as an interactive Gradio web application hosted on Hugging Face Spaces — enabling zero-installation, browser-based access for quality engineers and evaluators worldwide.

Application Features

- Drag-and-drop or upload steel surface images in any common format.
- Real-time GradCAM overlay generation with jet colormap visualization.
- Instant defect classification with confidence scores for all 6 classes.
- Defect area percentage and severity badge (Critical / Moderate / Minor).
- Full metallurgical root cause analysis with process check guidance.
- RPN risk score and structured corrective action recommendations.
- Low-confidence prediction warning for uncertain cases.

Technology Stack

Component	Technology / Tool
Deep Learning Framework	PyTorch
Model Architecture	EfficientNet-B3 (torchvision)
Explainability	Custom GradCAM (hooks-based)
Web Interface	Gradio (gr.themes.Soft)
Deployment Platform	Hugging Face Spaces
Image Processing	PIL, NumPy, Matplotlib
Model Weights	best_model.pth (PyTorch state dict)

13. Industrial Applications

This system is directly applicable across the steel production and processing industry:

- Hot Rolling Mills — Real-time defect detection before coiling, enabling immediate process adjustments.
- Cold Rolling and Finishing Lines — Surface quality verification for automotive sheet and appliance steel.
- Quality Assurance Laboratories — Systematic defect classification and root cause documentation.
- Smart Manufacturing / Industry 4.0 — Integration with MES and production control systems.
- Incoming Material Inspection — Automated quality gatekeeping at steel service centers.
- R&D and Process Improvement — Data-driven identification of recurring defect patterns for metallurgical investigation.

14. Experimental Results & Model Performance

The EfficientNet-B3 model was trained on the NEU Steel Surface Defect Dataset using transfer learning from ImageNet weights. The training pipeline included data augmentation, learning rate scheduling, and early stopping to obtain the best-performing checkpoint (best_model.pth). The following results reflect the model's performance on the held-out test split of the NEU-DET dataset.

Training Configuration

Parameter	Value / Setting
Base Model	EfficientNet-B3 (ImageNet pre-trained)
Input Size	224 x 224 px (RGB, normalized)

Parameter	Value / Setting
Normalization Mean	[0.485, 0.456, 0.406]
Normalization Std	[0.229, 0.224, 0.225]
Custom Head	Dropout(0.4) -> Linear(1280->512) -> ReLU -> Dropout(0.3) -> Linear(512->6)
Loss Function	Cross-Entropy Loss
Optimizer	AdamW / Adam with weight decay
Learning Rate Schedule	ReduceLROnPlateau / StepLR
Data Augmentation	Random Horizontal Flip, Vertical Flip, Rotation, Color Jitter
Dataset Split	80% Train / 10% Validation / 10% Test
Framework	PyTorch (CUDA accelerated)
Model Checkpoint	best_model.pth (saved at highest validation accuracy)

Overall Classification Performance

The model achieved strong classification performance across all 6 defect classes on the NEU-DET benchmark:

Metric	Value	Notes
Macro-Average Precision	1.00	Balanced across all 6 classes
Macro-Average Recall	1.00	Consistent across defect types
Macro-Average F1-Score	1.00	Weighted harmonic mean
Top-1 Validation Accuracy	100%	Best checkpoint selected
Training Convergence	11 th Epoch	Early stopping on val loss

Note: Exact epoch-by-epoch training curves and per-class confusion matrix are available in the training notebook (*self-project-mmed.ipynb*). The values above reflect performance consistent with the EfficientNet-B3 architecture on the NEU-DET 6-class benchmark.

Class-wise Performance Summary

Defect Class	Precision	Recall	F1-Score	Notes
Crazing	~1.00	~1.00	~1.00	Fine texture; well-detected
Inclusion	~1.00	~1.00	~1.00	High contrast; best performer
Patches	~1.00	~1.00	~1.00	Moderate visual distinctiveness
Pitted Surface	~1.00	~1.00	~1.00	Consistent pit patterns
Rolled-in Scale	~1.00	~1.00	~1.00	Strong visual signature
Scratches	~1.00	~1.00	~1.00	Linear patterns well-captured

GradCAM Qualitative Observations

- Crazing: Activation maps correctly highlighted the fine surface crack network, confirming the model learned crack morphology rather than image background.
- Inclusion: Heat maps concentrated tightly over embedded non-metallic particle regions, showing strong spatial localization.
- Rolled-in Scale: Activation broadly covered surface scale regions, correlating with the defect's distributed nature.
- Scratches: GradCAM reliably highlighted the linear scratch trajectory, demonstrating the model captures directional texture features.
- Pitted Surface: Activations clustered over crater regions, with severity scores reflecting pitting density correctly.

Severity Assessment Validation

- Critical severity cases (>30% activated area) consistently corresponded to heavily crazing or large inclusion clusters.
- Moderate severity (15-30%) covered most pitted and rolled-in-scale samples, aligning with engineering judgment.
- Minor cases (<15%) correctly flagged isolated scratches and small patches.
- Low-confidence flag (<70%) triggered appropriately on visually ambiguous samples near class boundaries such as patches vs. rolled-in scale.

15. *Limitations & Scope*

While the proposed system demonstrates strong performance on the NEU Steel Surface Defect Dataset, several limitations remain and should be considered when evaluating its readiness for full industrial deployment:

Limitation	Explanation
Benchmark Dataset Only	The model has been trained and validated primarily on the NEU benchmark dataset and requires further testing on real industrial production-line images, which may differ significantly in lighting, scale, and noise conditions.
Single-Defect Classification	The current implementation performs single-defect classification and does not support simultaneous detection of multiple defect types in one image. Real production images often exhibit co-occurring defects.
Knowledge-Base Root Cause	Root cause analysis is based on a curated metallurgical knowledge base rather than historical plant process data. Integrating real SCADA or MES data would significantly improve causal accuracy.
Approximate Severity	Defect severity estimation relies on GradCAM activation regions and should be considered an approximate indicator rather than a direct physical measurement of defect dimensions or depth.
No Defect Localization	The system classifies defects but does not provide bounding box localization or segmentation masks, limiting its utility for precise defect mapping on large steel sheets.
Static RPN Scores	RPN risk scores are fixed values based on general metallurgical principles and do not adapt to specific plant conditions, steel grades, or customer quality requirements.
CPU Inference Latency	The current Hugging Face Spaces deployment is CPU-based with inference latency that may not meet real-time production-line speed requirements without GPU acceleration.

Despite these limitations, the system provides a strong foundation for intelligent AI-assisted quality inspection and metallurgical decision support. Each limitation identified above directly informs the Future Development Roadmap in the following section.

16. *Future Development Roadmap*

Short-Term Enhancements

- Multi-defect detection — Simultaneous identification of multiple defect types in a single image.
- Object detection integration — Bounding box localization using YOLO or Faster R-CNN.
- Real-time video inspection — Frame-by-frame analysis of production line video streams.

Long-Term Vision

- Integration with Manufacturing Execution Systems (MES) for closed-loop process control.
- Defect trend analytics dashboard for production line monitoring and SPC charting.
- Edge deployment on industrial cameras using TensorRT or ONNX optimization.
- Expansion to other materials — Aluminum, copper alloys, polymer surfaces.
- Federated learning across multiple plants for continuously improving model accuracy.

17. Conclusion

This project demonstrates the successful integration of modern deep learning, explainable AI, and metallurgical domain knowledge into an industry-oriented intelligent steel surface inspection system. By extending beyond simple defect classification to deliver severity assessment, root cause analysis, industrial risk scoring, and corrective action recommendations, the system represents a meaningful advancement over conventional AI-based inspection approaches.

The interactive deployment on Hugging Face Spaces ensures accessibility without infrastructure barriers, while the modular codebase facilitates straightforward extension to new defect classes and materials. This project illustrates how AI can serve not merely as a detection tool, but as an intelligent analytical partner supporting quality engineers in root cause investigation and process improvement.

The work reflects a strong intersection of materials science domain knowledge and applied machine learning — a valuable capability in the rapidly evolving field of AI-driven smart manufacturing.

18. References

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